EXERCISE – 5.1

INFERENCE ABOUT POPULATION MEAN WHEN SIGMA IS KNOWN

1. Assume the population is normally distributed. Test the claim about the population mean μ at the level of significance α using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

(i) $n=30,\ \bar{x}=15.2,\ \sigma=3,\ \alpha=5\%,\ H_0:\mu=15.8,\ H_1:\mu\neq 15.8$

$(Ans: z=-1.095,\ P-value=0.2758,\ CI:(14.13\le\mu\le 16.27),\ Conclusion:\ do\ not\ reject\ H_0)$

(ii) $n=10,\ \bar{x}=27,\ \sigma=1.2,\ \alpha=1\%,\ H_0:\mu\le 26.3,\ H_1:\mu>26.3$

$(Ans: z=1.845,\ P-value=0.0325,\ CI:(\mu\ge 26.11),\ Conclusion: do\ not\ reject\ H_0)$

(iii) $n=49,\ \bar{x}=125,\ \sigma=4.2,\ \alpha=10\%\ \ H_0:\mu\ge 123.5,\ H_1:\mu<123.5$

$(Ans:\ z=2.5,\ \ P-value=0.9937,\ CI:(\mu\le 125.77),\ Conclusion: do\ not\ reject\ H_0)$

2. Assume the population is normally distributed. Test the claim about the population mean μ at the level of significance α using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

(i) $Claim:\ \mu=40,\ \ \alpha=0.05,\ \sigma=1.97,\ Sample\ Statistics:\ \bar{x}=39.2,\ n=25$

$(Ans: z=-2.03.,\ P-value=0.0424,\ CI:(38.43.13\le\mu\le 39.97), Conclusion:\ reject\ H_0)$

(ii) $Claim:\ \mu>10,\ \ \alpha=0.01,\ \sigma=3,\ \ Sample\ Statistics:\ \bar{x}=12,\ \ n=36$

$(Ans: z=4,\ \ P-value=0.000031,\ CI:(\mu\ge 10.84),\ Conclusion: reject\ H_0)$

(iii) $Claim:\ \mu\ge 1475,\ \ \alpha=0.07,\ \sigma=29,\ \ Sample\ Statistics:\ \bar{x}=1468,\ \ n=26$

$(Ans: z=-1.23,\ P-value=0.1093,\ \ CI:(\mu\le 1476.42),\ \ Conclusion: do\ not\ reject\ H_0)$

(iv) $Claim:\ \mu\le 22500,\ \alpha=0.01,\ \sigma=1200,\ Sample\ Statistics:\ \bar{x}=23500,\ n=45$

$(Ans: z=5.59,\ \ P-value=0.0000,\ \ \ CI:(\mu\ge 23083.19),\ Conclusion:\ reject\ H_0)$

(v) $Claim:\ \mu<220,\ \ \alpha=0.1,\ \ \sigma=15,\ \ \ Sample\ Statistics:\ \bar{x}=215,\ \ \ n=64$

$(Ans: z=-2.67,\ P-value=0.0038,\ \ \ CI:(\mu\le 217.4),\ Conclusion:\ reject\ H_0)$

(vi) $Claim:\ \mu\neq 5880,\ \ \alpha=0.03,\ \ \sigma=413,\ \ Sample\ Statistics:\ \ \bar{x}=5771,\ \ n=67$

$(Ans: z=-2.16,\ P-value=0.9692,\ \ \ CI:(5661.51\le\mu\le 5880.49), Conclusion:\ do\ not\ reject\ H_0)$

3. The mean lifetime of electric light bulbs produced by a company has in the past been 1120 hours with a standard deviation of 125 hours. A sample of 100 electric bulbs recently chosen from a supply of newly produced bulbs showed a mean lifetime 0f 1070 hours. Company claimed that lifetime of bulbs has not changed.

Test the claim at 5% level of significance using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

$(Ans: H_0:\mu=1120\ \ H_1:\mu\neq 1120,\ z=-4,\ P-value=0.0002, CI:(1045.5\le\mu\le 1094.5),\ Conclusion: reject\ H_0)$

4. A random sample of 100 medical school applicants at a university has a mean total score of 502 on the MCAT. According to a report, the mean total score for the school's applicants is more than 499. Assume the population standard deviation is 10.6. At $\alpha=0.01$, is there enough evidence to support the report's claim? Test the claim at 1% level of significance using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

$(Ans: H_0:\mu\le 499,\ \ H_1:\mu>499,\ z=2.83, P-value=0.0023,\ \ CI:(\mu\ge 499..53), Conclusion: reject\ H_0)$

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5. The mean salary of federal government employees on the General Schedule is \$59,593. The average salary of 30 state employees who do similar work is \$58,800 with $\sigma = \$1500$. At the 0.01 level of significance, can it be concluded that state employees earn on average less than federal employees? Test the claim at 1% level of significance using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

$(Ans: H_0: \mu \geq 59593,\ H_1: \mu < 59593,\ z = -2.89,\ P-value = 0.0019,\ CI: (\mu \leq 59438.09),\ Conclusion: reject\ H_0)$

6. An LED lamp manufacturer guarantees that the mean life of a certain type of LED lamp is at least 25,000 hours. A random sample of 49 LED lamps has a mean life of 24,800 hours. Assume the population is normally distributed and the population standard deviation is 500 hours. At $\alpha = 0.05$, do you have enough evidence to reject the manufacturer's claim? Test the claim at 5% level of significance using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

$(Ans: H_0: \mu \geq 25000,\ H_1: \mu < 25000,\ z = -2.8,\ P-value = 0.0026,\ CI: (\mu \leq 24917.86),\ Conclusion: \text{Re}\ ject\ H_0)$

7. A random sample of 9 values from a normal population gave the values $66, 68, 59, 67, 64, 66, 63, 62, 61$. Discuss the suggestion that the mean in the population is 60, given that population standard deviation is 3 i.e. $\sigma = 3$. At 5% level of significance Test the hypothesis using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

$(Ans: H_0: \mu = 60,\ H_1: \mu \neq 60,\ z = 4,\ CI: (62.04 \leq \mu \leq 65.96),\ P-value = 0.0002,\ Conclusion: reject\ H_0)$

8. From the previous records it is known that the duration of treating a disease by a standard medicine has a mean of 15 days and a standard deviation of 3 days. It is claimed that a new medicine can reduce the treatment time, to test this claim, the new medicine is tried on 70 patients and their mean time of recovery is recorded to be 14.5 days. Test the claim at 5% level of significance using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

$(Ans\ H_0: \mu \geq 15, H_1: \mu < 15,\ z = -1.394,\ CI: (\mu \leq 15.09),\ P-value = 0.0816,\ Conclusion: do\ not\ reject\ H_0)$

9. The breaking strength of cables produced by a company has a mean of 1800 pounds and standard deviation of 100 pounds. By a new technique in the production process, it is claimed that the breaking strength can be increased. To test this claim, a sample of 50 cables is tested and it is found that the mean breaking strength is 1850 pounds. Test the claim at 1% level of significance using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

$(Ans\ H_0: \mu \leq 1800,\ H_1: \mu > 1800,\ z = 3.54,\ CI: (\mu \geq 1817.04),\ P-value = 0.0002,\ Conclusion: reject\ H_0)$

10. A recent national survey found that high school students watched an average of 6.8 movies per month with a population standard deviation of 1.8. The distribution of number of movies watched per month follows the normal distribution. A random sample of 36 college students revealed that the mean number of movies watched last month was 6.2. At the 0.05 significance level, can we conclude that college students watch fewer movies a month than high school students? Can we support the claim at 5% level of significance using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

$(Ans\ H_0: \mu \geq 6.8,\ H_1: \mu < 6.8,\ z = -2,\ CI: (\mu \leq 5.705),\ P-value = 0.0228,\ Conclusion: reject\ H_0)$

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EXERCISE – 5.2

INFERENCE ABOUT POPULATION MEAN WHEN SIGMA IS UNKNOWN

1. Assume the population is normally distributed. Test the claim about the population mean μ at the level of significance α using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

(i) $H_0: \mu = 1503,\ H_1 = \mu \neq 1503,\ \alpha = 10\%,$ *Sample Statistics* $: \bar{x} = 1505.8,\ s = 3.1874,\ n = 6$

$(Ans: t = 2.15,\ P-value = 0.0842,\ CI: (1503.18 \leq \mu \leq 1508.42)$, *Conclusion*: *reject* $H_0)$

(ii) $H_0: \mu \geq 133,\ H_1 = \mu < 133,\ \alpha = 1\%,$ *Sample Statistics*: $\bar{x} = 129.8,\ s = 3.29,\ n = 10$

$(Ans: t = -3.07,\ P-value = 0.9934,\ CI: (\mu \leq 132.73),$ *Conclusion*: *do not reject* $H_0)$

(iii) $H_0: \mu \leq 24.1,\ H_1: \mu > 24.1,\ \alpha = 5\%,$ *Sample Statistics*: $\bar{x} = 24.9,\ s = 0.775,\ n = 12$

$(Ans: t = 3.58,\ P-value = 0.0000,\ CI: (\mu \geq 24.50),$ *Conclusion*: *reject* $H_0)$

2. Assume the population is normally distributed. Test the claim about the population mean μ at the level of significance α using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

(i) *Claim*: $\mu = 15,\ \alpha = 0.01,$ *Sample Statistics* $: \bar{x} = 13.9,\ s = 3.23,\ n = 16$

$(Ans: t = -1.36,\ P-value = 0.1932,\ CI: (11.52 \leq \mu \leq 16.28),$ *Conclusion*: *do not reject* $H_0)$

(ii) *Claim*: $\mu > 25,\ \alpha = 0.05,$ *Sample Statistics* $: \bar{x} = 26.2,\ s = 2.32,\ n = 17$

$(Ans: t = 2.13,\ P-value = 0.0244,\ CI: (\mu \geq 16.28),$ *Conclusion*: *reject* $H_0)$

(iii) *Claim*: $\mu \geq 8000,\ \alpha = 0.01,$ *Sample Statistics*: $\bar{x} = 7700,\ s = 450,\ n = 25$

$(Ans: t = -3.33,\ P-value = 0.0013, CI: (\mu \geq 16.28),$ *Conclusion*: *reject* $H_0)$

(iv) *Claim*: $\mu \leq 1600,\ \alpha = 0.10,$ *Sample Statistics* $: \bar{x} = 1550,\ s = 165,\ n = 26$

$(Ans: t = -1.54,\ P-value = 0.9322, CI: (\mu \leq 16.28),$ *Conclusion*: *do not reject* $H_0)$

(v) *Claim*: $\mu < 4915,\ \alpha = 0.01,$ *Sample Statistics* $: \bar{x} = 5017,\ s = 5613,\ n = 11$

$(Ans: t = 0.06,\ P-value = 0.5234, CI: (\mu \leq 16.28),$ *Conclusion*: *do not reject* $H_0)$

(vi) *Claim*: $\mu \neq 52200,\ \alpha = 0.05,$ *Sample Statistics*: $\bar{x} = 53220,\ s = 2700,\ n = 34$

$(Ans: t = 2.20,\ P-value = 0.0347, CI: (52277.92 \leq \mu \leq 54162.07),$ *Conclusion*: *reject* $H_0)$

3. *The average local cell phone call length was reported to be 2.27 minutes. A random sample of 20 phone calls showed an average of 2.98 minutes in length with a standard deviation of 0.98 minute. At* a *0.05 can it be concluded that the average differs from the population average?* Test the claim for a given level of significance α using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

$(Ans\ H_0: \mu = 2.27,\ H_1: \mu \neq 2.27,\ t = 3.24,\ CI: (2.521 \leq \mu \leq 3.439),\ P-value = 0.0043,$ *Conclusion*: *reject* $H_0)$

4. *A survey of 15 large U.S. cities finds that the average commute time one way is 25.4 minutes. A chamber of commerce executive feels that the commute in his city is less and wants to publicize this. He randomly selects 25 commuters and finds the average is 22.1 minutes with a standard deviation of 5.3 minutes. At* $\alpha = 0.10$ *, is he correct?* Test the claim using

(a) *Critical value Approach* (b) $P-$*value Approach* (c) *Confidence Interval Approach*

$(Ans\ H_0: \mu \geq 25.4,\ H_1: \mu < 25.4,\ t = -3.11,\ CI: (\mu \leq 23.50), P-value = 0.0024,$ *Conclusion*: *reject* $H_0)$

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5. *During a recent year the average cost of making a movie was $54.8 million. This year, a random sample of 15 recent action movies had an average production cost of $62.3 million with a variance of $90.25 million. At the 0.05 level of significance, can it be concluded that it costs more than average to produce an action movie?* Test the claim for a given level of significance α using

$(a)\,Critical\,value\ Approach\ (b)\ P-value\ Approach\ \ (c)\,Confidence\ Interval\ Approach$

$(Ans\ \ H_0:\mu\le 54.8,\ \ H_1:\mu>54.8,\ t=3.06,\ CI:(\mu\ge 66.62),\ P-value=0.0043,\ Conclusion:\ reject\ H_0)$

6. A call center claims that the mean wait time for a customer to connect with a customer care executive is at most 2.5 minutes. A random sample of 35 calls at the call center has a mean wait time of 2.75 minutes and a standard deviation of 1.25 minutes. Is there enough evidence to reject the claim at $\alpha=0.05$? Test the claim for a given level of significance α using

$(a)\,Critical\,value\ Approach\ (b)\ P-value\ Approach\ \ (c)\,Confidence\ Interval\ Approach$

$(Ans\ \ H_0:\mu\le 2.5,\ \ H_1:\mu>2.5,\ t=1.1832,\ CI:(\mu\ge\ \ \ \ ),\ P-value=0.1225,\ Conclusion:do\ not\ reject\ H_0)$

7. The mean income per person in the United States is $50,000, and the distribution of incomes follows a normal distribution. A random sample of 10 residents of Wilmington, Delaware, had a mean of $60,000 with a standard deviation of $10,000. Check whether residents of Wilmington, Delaware, have more income than the national average? Using

$(a)\,Critical\,value\ Approach\ (b)\ P-value\ Approach\ \ (c)\,Confidence\ Interval\ Approach$

$(Ans\ \ H_0:\mu\le 50000,\ \ H_1:\mu>50000,\ t=3.1622,\ CI:(\mu\ge\ \ \ \ ),\ \ P-value=0.0058,\ Conclusion:\mathrm{Re}\ ject\ H_0)$

8. The Rocky Mountain district sales manager of Rath Publishing Inc., a college textbook publishing company, claims that the sales representatives make an average of 40 sales calls per week on professors. Several reps say that this estimate is too low. To investigate, a random sample of 28 sales representatives reveals that the mean number of calls made last week was 42. The standard deviation of the sample is 2.1 calls. Using the 0.05 significance level, can we conclude that the mean number of calls per salesperson per week is more than 40? Check the claim using

$(a)\,Critical\,value\ Approach\ (b)\ P-value\ Approach\ \ (c)\,Confidence\ Interval\ Approach$

$(Ans\ \ H_0:\mu\le 40,\ \ H_1:\mu>40,\ t=5.04,\ CI:(\mu\ge\ \ \ \ ),\ \ P-value=0.0000,\ Conclusion:\mathrm{Re}\ ject\ H_0)$

9. The Cholesterol levels of 8 women were measured, with the following results

$$3.1,\ 2.8,\ 1.5,\ 1.7,\ 2.4,\ 1.9,\ 3.3,\ 1.6.$$

Test at 5% level of significance, whether the sample has been drawn from a distribution with mean cholesterol level 3.1. using

$(a)\,Critical\,value\ Approach\ (b)\ P-value\ Approach\ \ (c)\,Confidence\ Interval\ Approach$

$(Ans\ \ H_0:\mu=3.1,\ \ H_1:\mu\ne 3.1,\ t=-3.23,\ CI:(1.69\le\mu\le 2.88),\ \ P-value=0.0121,\ Conclusion:\ reject\ H_0)$

10. An athlete finds that her times for running a race are normally distributed with mean 10.6 seconds. She trains intensively for a week and then records her time in the next 6 races. Her times, in seconds, are

$$10.70,\ 10.65,\ 10.75,\ 10.80,\ 10.60,10.55.$$

It is claimed that training has improved her times. Check the claim at $\alpha=0.05$ using

$(a)\,Critical\,value\ Approach\ (b)\ P-value\ Approach\ \ (c)\,Confidence\ Interval\ Approach$

$(Ans\ \ H_0:\mu\ge 10.6,\ \ H_1:\mu<10.6,\ t=1.9640,\ CI:(\mu\le\ \ \ \ ),\ \ P-value=0.9466,\ Conclusion:\ do\ not\ reject\ H_0)$

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